

Statistik_2022-05-18

Can the function

$$p(x) = \begin{cases} ax^2 + 2x - 1 & x = 1, 2, 3 \\ 0 & \text{else} \end{cases}$$

be the probability mass function for some discrete random variable? Here a is a real number.

- ☐ a. Yes, only for a unique negative a .
- ☐ b. No, because probabilities cannot be negative.
- ☐ c. Yes, only for a unique positive a .
- ☐ d. No, because probabilities cannot be greater than 1.

$$(a1^2 + 2 \cdot 1 - 1) + (a2^2 + 2 \cdot 2 - 1) + (a3^2 + 2 \cdot 3 - 1) = 13a + 12 - 3 = \boxed{13a + 9}$$

$$13a + 9 = 1$$

$$13a + 8 = 0$$

$$a = -\frac{8}{13}$$

Einsetzen:

$$-\frac{8}{13} \cdot x^2 + 2x - 1$$

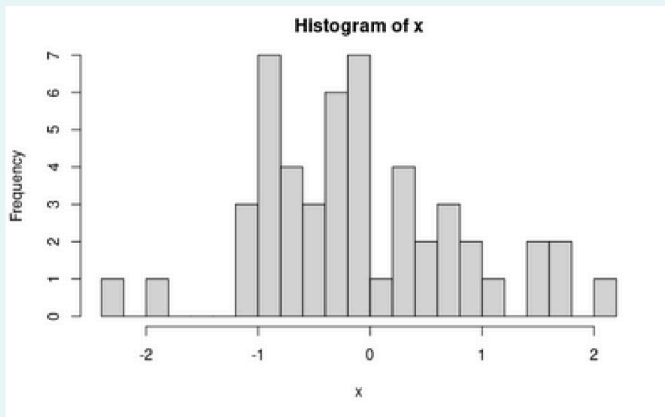
$$-\frac{8}{13} \cdot 1^2 + 2 \cdot 1 - 1 = 0.38$$

$$-\frac{8}{13} \cdot 2^2 + 4 - 1 = 0.53$$

$$-\frac{8}{13} \cdot 3^2 + 2 \cdot 3 - 1 = -0.53 \dots$$

Kann nicht negativ sein $\implies b)$

Consider the following histogram of a data set consisting of 50 observations. Which of the following statements is **true**.



- ☐ a. The standard deviation of these data is at least 1.5.
- ☐ b. The data are approximately normal.
- ☐ c. The sample mean is a bad summary of these data.
- ☐ d. The data are asymmetric.

$\Rightarrow d)$

A 95% confidence interval from a poll performed to estimate the population proportion of eligible voters who would not vote in the next Presidential election is (40%, 50%). Which of the following statements is **true**?

- ☐ a. The sample size that was used in the poll was approximately 1,050.
- ☐ b. If the poll were to be repeated in an identical fashion then there is a 95% chance that the new sample proportion will lie in the range 40% to 50%.
- ☐ c. A 90% confidence interval would have been narrower.
- ☐ d. The probability that the population proportion lies in this interval is 0.95.

⇒ c)

Suppose you toss a fair coin ten times and it comes up heads every time. Which one of the following statements is **true**?

- ☐ a. The probability that the next toss will again be heads is 0.5.
- ☐ b. By the law of large numbers, the next toss is more likely to be tails than another heads.
- ☐ c. By the properties of conditional probability, the next toss is more likely to be heads given that ten tosses in a row have been heads.
- ☐ d. The law of large numbers tells how many tosses will be necessary before the percentages of heads and tails are again in balance.

⇒ a)

Bieber Fever (i.e., being a huge Justin Bieber fan) is a growing health concern among America's youth. To investigate it, I gave a survey to 200 middle school students in New York City which asked the following two questions: "What is your gender?" and "Do you have Bieber Fever?" In my data set, 40% of respondents have Bieber Fever, 60% of the respondents were girls, and among the boys, 15% have Bieber Fever.

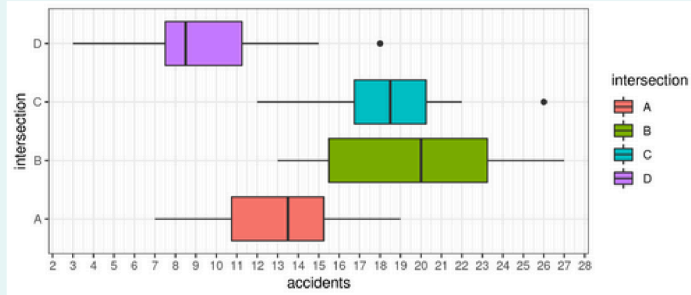
Under the null hypothesis of independence, how many boys do we expect to have Bieber Fever?

- ☐ a. 91
- ☐ b. 49
- ☐ c. 36
- ☐ d. 32

$$\begin{aligned}
 n &= 200 \\
 P(Girl) &= 0.6 \\
 P(Boy) &= 0.4 \\
 P(BF) &= 0.4 \\
 P(B|BF) &= 0.15
 \end{aligned}$$

$$200 \cdot 0.4 \cdot 0.4 = \boxed{32}$$

Data on the number of yearly accidents were collected from four intersections (A-D) over a 20 year period. Which of the following statements is **false**?



- ☐ a. The minimum number accidents that occurred in a single intersection was 3.
- ☐ b. During at least 5 years, fewer than 10 accidents occurred at intersection A.
- ☐ c. The minimum accident total at intersection C was higher than the number of accidents observed at intersection D in 75% of years.
- ☐ d. During at least 75% of years, intersection C had more accidents than the lowest 75% of years at intersection A.

b weil mans bei einem boxplot nicht erkennen kann...

Let $X_1, \dots, X_{100}$ be a random sample from an exponential distribution with the expectation $\frac{1}{2}$. Determine the approximate value of

$$P\left(\sum_{i=1}^{100} X_i > 53\right)$$

using the Central limit theorem.

Recall, a random variable with probability density function

$$f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & x \leq 0 \end{cases}$$

is exponential with parameter $\lambda = 2$ and its expectation equals $\frac{1}{2}$.

Use the values given in the table below.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051

Table 1: Cumulative distribution function of the standard normal distribution

- ☐ a. 0.345
- ☐ b. 0.309
- ☐ c. 0.271
- ☐ d. 0.274

Nach dem Zentralen Grenzwertsatz ist die Summe $S_n = \sum X_i$ für große n näherungsweise normalverteilt:

$$S_n \approx \mathcal{N}(n\mu, n\sigma^2)$$

Für $X \sim \text{Exp}(\lambda)$ gilt:

- $E(X) = \frac{1}{\lambda}$
- $\text{Var}(X) = \frac{1}{\lambda^2}$

1. Kenngrößen berechnen

- $n = 100$
- $\mu = \frac{1}{2} = 0.5$
- $\sigma^2 = \frac{1}{4} = 0.25$
- $\mu_S = 100 \cdot 0.5 = 50$ (für die Summe)
- $\sigma_S = \sqrt{100 \cdot 0.25} = 5$

2. Standardisierung (Z-Score)

Um die Standardnormalverteilungstabelle zu nutzen, wird der Wert $x = 53$ transformiert:

$$z = \frac{x - \mu_S}{\sigma_S} = \frac{53 - 50}{5} = 0.6$$

3. Wahrscheinlichkeit bestimmen

$$\begin{aligned}P(S_{100} > 53) &\approx P(Z > 0.6) \\&= 1 - \Phi(0.6) \\&= 1 - 0.7257 \quad (\text{laut Tabelle}) \\&= 0.2743\end{aligned}$$

Ergebnis: $P \approx \boxed{0.274}$

I have data on the daily high temperature (in Celsius) in Vienna and the daily total revenue from ice cream sales at Eis Greissler locations in Vienna (in thousands of Euros). The relationship is linear, and the correlation between the two variables is $r = 0.4$.

For a day in which the high temperature was 2 standard deviation above the mean, what would be our best estimate for the standardized value of ice cream sales at Eis Greissler?

- ☐ a. 0.4
- ☐ b. We cannot say without knowing the estimated slope, b_1 .
- ☐ c. We cannot say without knowing the standard deviations of temperature and revenue.
- ☐ d. 0.8

$$z_{\hat{y}} = r \cdot z_x$$

$$z_{\hat{y}} = 0.4 \cdot 2 = 0.8$$

Assume the given distributions are normal. Pumpkins grown on a certain farm have weights with a mean of 10 kilograms. What is the standard deviation (in kilograms) of the weight if 37% of the pumpkins weigh more than 13 kilograms?

The values given in the table below should be used.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026
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0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051

Table 1: Cumulative distribution function of the standard normal distribution

- ☐ a. 8.82
- ☐ b. 2.97
- ☐ c. 1.22
- ☐ d. 4.76

1. Gegebene Werte und Ziel

- Verteilung: Normalverteilung mit $\mu = 10$ und gesuchtem σ .
- Bedingung: $P(X > 13) = 0,37$.
- Daraus folgt für die kumulative Verteilungsfunktion: $P(X \leq 13) = 1 - 0,37 = 0,63$.

2. Standardisierung (Z-Transformation)

Wir transformieren X in die Standardnormalverteilung Z :

$$P\left(Z \leq \frac{13 - 10}{\sigma}\right) = 0,63$$

$$\Phi\left(\frac{3}{\sigma}\right) = 0,63$$

3. Tabellenwert finden

In der bereitgestellten Tabelle suchst du den Wert, der 0,63 am nächsten kommt:

- In Zeile **0.3**, Spalte **0.03** steht 0,6293.
- In Zeile **0.3**, Spalte **0.04** steht 0,6331.

Obwohl 0,6293 näher an 0,63 liegt, führt die Verwendung von $z = 0,34$ (was 0,6331 entspricht) exakt zu einer der Antwortmöglichkeiten.

4. Nach σ auflösen

Setze den gefundenen z-Wert 0,34 gleich dem Argument der Φ -Funktion:

$$\begin{aligned}\frac{3}{\sigma} &= 0,34 \\ \sigma &= \frac{3}{0,34} \\ \sigma &\approx 8,8235\end{aligned}$$

Der Wert gerundet ist **8,82**. Die richtige Antwort ist somit **a**.

Which of the following holds **true** for two independent random variables X and Y ?

- ☐ a. $\text{Var}(X + Y) \leq \text{Var}(Y)$
- ☐ b. $\text{Var}(2X - Y) = 2\text{Var}(X) - \text{Var}(Y)$
- ☐ c. $\text{Var}(2X - Y) \leq 4\text{Var}(X)$
- ☐ d. $\text{Var}(X + Y) \geq \text{Var}(X)$

d weil

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

I have collected data from international students from 5 different European countries. I asked whether or not they smoke and what country they are from. I am interested in testing whether or not the proportion that smokes is different between countries but am concerned about the number of hypothesis tests this entails. Which of the following is **true**?

- ☐ a. I could do a chi-squared test for goodness of fit, where I suppose that each country has a proportion of smokers equal to the proportion of smokers I've observed in my entire data set (thus giving me 4 degrees of freedom).
- ☐ b. Only the ANOVA provides a valid test to compare all countries.
- ☐ c. It doesn't matter whether I use an ANOVA or correct the individual tests via the Bonferroni correction as they both have the same assumptions.
- ☐ d. I could test whether each country has 50% smokers, and adjust these 5 tests using the Bonferroni correction.

a weil

$$df = (5 - 1) \times (2 - 1) = 4 \times 1 = 4$$

A drug company would like to show that their new drug is better than the currently used one (the status-quo). In the hypothesis testing framework if they make a Type II error then what is the practical consequence of the error?

- ☐ a. Without data from an actual hypothesis test, it is not possible to interpret the consequence of the Type II error.
- ☐ b. They correctly claim that their drug is better when it really is better.
- ☐ c. They incorrectly claim that their drug is better when really it is not.
- ☐ d. They incorrectly conclude that their drug is no better when really it is.

d

Which one of the following statements is **true**?

- ☐ a. If $A^c \cap B^c = \emptyset$ then $P(A \cap B) = P(A) + P(B)$
- ☐ b. $P(A \cap B)$ may be larger than $P(B)$.
- ☐ c. If $A^c \cap B^c = \emptyset$ then $P(A \cup B) < 1$
- ☐ d. If $A^c \cap B = \emptyset$ then $P(B) \leq P(A)$.

d) weil Alle Elemente von B müssen auch in A liegen.

A scientist tests 1000 hypotheses over his lifetime. Since he understands power well, he makes sure that each experiment is evaluated with a test that has exactly 80% power against a specific alternative hypothesis. He follows the usual tradition of doing tests at the 5% significance level. Suppose that, among all of the test that he performs in his lifetime, the null is actually true in 10% of the cases, while the specific alternative holds otherwise. In total, how many null hypotheses will he expect to reject?

- ☐ a. 770
- ☐ b. 545
- ☐ c. 650
- ☐ d. 725

1. Aufteilung der 1000 Tests

- H_0 ist wahr (Null is true): 10% der Fälle.

$$1000 \cdot 0,10 = 100 \text{ Fälle}$$

- H_1 ist wahr (Alternative holds): Die restlichen 90%.

$$1000 \cdot 0,90 = 900 \text{ Fälle}$$

2. Berechnung der Ablehnungen (Rejections)

Wir lehnen H_0 in zwei Situationen ab: Fälschlicherweise (Fehler 1. Art) oder korrekterweise (Power).

- Gruppe 1: H_0 ist wahr (100 Fälle)

Hier lehnen wir nur ab, wenn wir einen Fehler machen. Die Wahrscheinlichkeit dafür ist das Signifikanzniveau $\alpha = 0,05$.

$$\text{Rejections} = 100 \cdot 0,05 = 5$$

- Gruppe 2: H_1 ist wahr (900 Fälle)

Hier lehnen wir ab, wenn der Test anschlägt. Die Wahrscheinlichkeit dafür ist die Power = 0,80.

$$\text{Rejections} = 900 \cdot 0,80 = 720$$

3. Gesamtsumme

$$5 + 720 = 725$$

A 95% confidence interval for a population mean based on a sample of size 400 was (20, 28). Which of the following is **true**?

- ☐ a. We cannot say whether or not the true mean is in (20, 28).
- ☐ b. If we collected a new sample of size 800, the resulting 95% confidence interval would have a higher probability of containing the true mean.
- ☐ c. There is a 95% probability that 24 is the true value of the population mean.
- ☐ d. There is a 95% probability that the true value of the population mean is in (20,28).

a) weil es kein confidence interval ist sondern einfach in Intervall indem der mean liegt...

Let $X \sim \mathcal{N}(1, 4)$. Express

$$P(X^2 \geq 2X)$$

in terms of the cumulative distribution function Φ of the standard normal random variable

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt, \quad x \in \mathbb{R}.$$

- ☐ a. $1.5 - \Phi(2)$
- ☐ b. $2 - 2 \cdot \Phi(0.5)$
- ☐ c. $2 - 2 \cdot \Phi(0.25)$
- ☐ d. $1 - \Phi(1)$

1. Ungleichung auflösen

Die Bedingung innerhalb der Wahrscheinlichkeit muss vereinfacht werden.

$$\begin{aligned} X^2 &\geq 2X \\ X^2 - 2X &\geq 0 \\ X(X - 2) &\geq 0 \end{aligned}$$

Damit das Produkt ≥ 0 ist, müssen wir die Bereiche außerhalb der Nullstellen betrachten.

Das ist erfüllt, wenn:

$$X \leq 0 \quad \text{oder} \quad X \geq 2$$

2. Standardisierung (Z-Score)

Wir wandeln X in die Standardnormalverteilung Z um:

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 1}{2}$$

Wir berechnen die Wahrscheinlichkeiten für beide Bereiche und nutzen die Symmetrieeigenschaft

$$\Phi(-z) = 1 - \Phi(z).$$

Linker Bereich ($X \leq 0$):

$$\begin{aligned} P(X \leq 0) &= P\left(Z \leq \frac{0 - 1}{2}\right) \\ &= P(Z \leq -0.5) \\ &= \Phi(-0.5) \\ &= 1 - \Phi(0.5) \end{aligned}$$

Rechter Bereich ($X \geq 2$):

$$\begin{aligned} P(X \geq 2) &= P\left(Z \geq \frac{2 - 1}{2}\right) \\ &= P(Z \geq 0.5) \\ &= 1 - P(Z \leq 0.5) \\ &= 1 - \Phi(0.5) \end{aligned}$$

3. Gesamtergebnis

Summe der beiden Wahrscheinlichkeiten:

$$P = (1 - \Phi(0.5)) + (1 - \Phi(0.5))$$

$$P = 2 - 2 \cdot \Phi(0.5)$$

An analyst has an iid sample of size 100, for which the sample mean equals 50 and the sample standard deviation is 30. Which of the following is an approximate 95% confidence interval for the population mean μ ? Use the normal quantiles below.

q_α	$q_{0.895}$	$q_{0.90}$	$q_{0.950}$	$q_{0.975}$	$q_{0.990}$	$q_{0.995}$	$q_{0.999}$
	1.253	1.280	1.645	1.960	2.33	2.58	3.09

- ☐ a. (43.73, 56.27)
- ☐ b. (45.89, 54.11)
- ☐ c. (49.37, 50.63)
- ☐ d. (44.12, 55.88)

$$x \pm z \left(1 - \frac{\alpha}{2}\right) \cdot \frac{s}{\sqrt{n}} = 50 \pm z \left(1 - \frac{0.05}{2}\right) \cdot \frac{30}{\sqrt{100}} = [44.12, 55.88]$$

A computer technician notes that 40% of computers fail because of the hard drive, 25% because of the monitor, 20% because of a disk drive, and 15% because of the microprocessor.

If the problem is not in the disk drive, what is the probability that it is in the microprocessor? (Note that only one failure occurs at a time.)

- ☐ a. 0.235
- ☐ b. 0.533
- ☐ c. 0.187
- ☐ d. 0.417

$$P(\text{Micro}|\text{Not Disk}) = \frac{0,15}{0,80} = 0,1875$$

Which of the following is the appropriate justification for using the paired t-test instead of a two-sample t-test? Assume the data are such that a paired t-test is possible.

- ☐ a. We hope to increase the standard error of the estimated difference.
- ☐ b. There are two samples whose means we wish to compare.
- ☐ c. We want to remove unwanted variability between subjects.
- ☐ d. By creating a new data set of differences, we are reducing the sample size. This increases the power of the test.

c)

Warum die anderen falsch sind

- a:** Wir wollen den Standardfehler verringern, nicht erhöhen. Ein höherer Standardfehler macht Ergebnisse ungenauer.
- b:** Das ist die Voraussetzung für *beide* Tests, aber kein Argument für den gepaarten Test.
- d:** Eine Reduktion der Stichprobengröße (Sample Size) ist eigentlich schlecht für die Power. Der gepaarte Test hat *trotz* weniger Freiheitsgraden oft mehr Power, aber nur wegen der Varianzreduktion, nicht wegen der kleineren Stichprobe.

In the context of a statistical test at significance level α , the p-value is below α . Which of the following is **false**?

- ☐ a. The test statistic lies in the rejection region.
- ☐ b. The value of the null hypothesis lies within the $(1 - \alpha)100\%$ confidence interval computed from these data.
- ☐ c. We would also reject at level 2α .
- ☐ d. The rejection region would be bigger if we would have used level 2α .

b)

- **Logik:** Es gibt eine direkte Verbindung zwischen Hypothesentests und Konfidenzintervallen.
 - Wenn wir H_0 **verwerfen** (was wir hier tun, da $p < \alpha$), dann liegt der Wert der Nullhypothese **außerhalb** des Konfidenzintervalls.
 - Wenn wir H_0 **behalten**, liegt der Wert **innerhalb** des Intervalls.
- Da wir hier verwerfen, darf der Wert nicht im Intervall liegen. Die Aussage behauptet aber, er liegt drin ("lies within").
- Die Aussage ist also **FALSCH**.